

Deposition kinetics of silicon dioxide from tetraethylorthosilicate by PECVD

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Abstract

Tetraethylorthosilicate (TEOS)-based SiO_2 films were prepared in an RF reactor at various discharge powers and two substrate temperatures, and infra-red absorption spectra were taken for each deposited film using an FTIR spectrometer. Based on the FTIR observations, a model for the plasma and surface kinetics was developed. Using a proposed three-step reaction mechanism at the surface of a two-dimensional lattice, the growth rate of the films and the incorporation rate of Si—O—Si bridges into the films were derived as functions of the densities of the film precursors, oxygen atoms and molecular oxygen ions. The densities are further given by the uniform discharge model of our previous work as a function of the discharge power. According to the model, in the regime of high ratio of oxygen to TEOS flow the growth rate is governed by the film precursors generated from TEOS, and the incorporation rate is additionally determined by oxygen radicals and molecular oxygen ions and by the dehydration process. The growth rate was measured in $\mu\text{m}/\text{min}$ and the relative incorporation rate was obtained from the absorption intensity of Si—O—Si rocking mode in FTIR spectra. The formulation of the model was compared with the experimental results, and on the validity of the model was discussed. © 2000 Elsevier Science S.A. All rights reserved.

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1. Introduction

Tetraethylorthosilicate (TEOS)-based SiO_2 has been extensively explored in a thermal or plasma assisted CVD system. Due to its Si—O bonds, TEOS can be dissociated to SiO_2 by pyrolysis or on the heated surface of a substrate [1]. Ozone is added as an additional process gas to take advantage of low temperature processes [2]. The plasma process can further lower the process temperature [3]. In a plasma process TEOS is introduced directly into the plasma with oxygen or indirectly showered in the downstream of oxygen plasma [4,5].

In such explorations, studies on the deposition mechanism have been of great interest as one of the efforts to obtain highly reliable dielectric materials (with high growth rate, low hydrogen or carbon content and high step coverage). Sorita et al. [6] attempted to explain the deposition mechanism for TEOS- SiO_2 films formed by pyrolysis using a mathematical model which considered the material balance for the reactant and intermediate species generated from reactant in the gas phase. According to the proposed deposition model, the formation of SiO_2 films is caused by two kinds of

intermediates, a high-activity one with a sticking probability near unity and a low-activity one with a sticking probability near zero, which undergo mutual reversible reactions in the gas phase in the high pressure regime. Hauptfear et al. [7] proposed a Langmuir–Hinshelwood type surface kinetics model in a cold wall deposition, in which TEOS is adsorbed in the surface site, where it is subsequently abstracted of a ligand by a vacant adjacent site. As a result, the site is occupied by hydrocarbons or oxygenates, which hinder further adsorption of TEOS and thus, limit the growth rate. In a kinetic study of the reaction of TEOS with oxygen atoms, Sanogo and Zachariah [8] proposed a gas kinetics model in which a two-step oxidation of TEOS with oxygen atoms leads to the formation of R_3SiO being $\text{R} = \text{C}_2\text{H}_5\text{O}$, which undergoes both elimination of ethylene and addition reactions to produce Si—O—Si bridges by dehydration.

The process of PECVD using TEOS/ O_2 is similar to that of pyrolysis, in that active film precursors are provided from the gas phase, similar to that of thermal CVD in that the substrate is heated and thus, some film precursors are dissociated on the surface, and to that of CVD using TEOS/ozone in that active oxygen radicals are supplied from the gas phase. In addition, ions are known to play a critical role in the process of PECVD. In a recent work on the modeling of high-density plasma deposition of SiO_2 from SiH_4 , O_2 and

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Ar gas mixtures by Meeks et al. [9], 167 gas-phase reactions and 96 surface reactions with the reaction kinetics parameters for each reaction are involved in the calculation. Since the electron impact dissociation and other key excitation cross-sections for TEOS are not known, the development of a rigorous deposition model for TEOS/O₂ system has hardly been explored. Raupp et al. [10] presented a simple model for TEOS/O₂ discharges which predicts trends in growth rate as functions of several deposition parameters. The model concentrated on the production and loss of oxygen atoms and molecular oxygen ions, but ignored the role of TEOS molecules and/or TEOS fragments. Some experimental findings, however, show that the growth rate increases or decreases with TEOS flow with fixed O₂ flow [3,5], which indicates that TEOS concentration or flux should be also taken into account for the modeling.

In a recent work on plasma polymerization, we focused on the dissociation of a chemical bond of monomer gas, not on chemical species, and presented a simple uniform plasma model, which allows to estimate ‘dissociation efficiency’, χ , in W⁻¹ of a chemical bond for a given plasma system, such as the Si—O—Si bridge in hexamethyldisiloxane (HMDSO) [11]. In a previous work related to deposition kinetics of silicon dioxide from hexamethyldisilazane (HMDSN)/O₂ discharges [12], we noted that χ for the dissociation of O₂ decreases with oxygen flow rate. With this estimated parameter the growth rate is predictable as a function of discharge power, flow rate, or pressure, even though the electron impact dissociation cross-sections are not known. In this work, we present a more advanced model for the deposition of SiO₂ from TEOS/O₂ discharges developed on the basis of the simple uniform discharge model [11] and the surface kinetics model [12]. While the models already studied focused on the deposition of stoichiometric SiO₂, which could affect growth rate prediction, the surface kinetics model of this work gives the growth rate and the incorporation rate of Si—O—Si bridges into the films separately, so that the proposed model can predict the integrity of the films, e.g. the content of OH bonds in the films.

2. Experiment

Films were deposited in a capacitively coupled parallel plate RF (13.56 MHz) discharge reactor. As the detailed scheme of the reactor and the deposition process are presented elsewhere [13], we give here a brief description. The two electrodes are aluminum plates with a spacing of 50 mm, the upper plate of Ø60 cm is grounded together with the reactor walls and the lower plate of Ø17 cm is powered. The reaction chamber was pumped to a background pressure of 1×10^{-5} Torr. Substrates, silicon wafers of 76 mm diameter, were put on the lower electrode, and when necessary, heated to a temperature of 200°C with a heating plate under the electrode. TEOS in the liquid state in a stainless steel vessel was evaporated through a liquid flow controller

and introduced into the reactor, and oxygen gas was introduced using a gas flow controller. Flow rate of the monomer and oxygen were held to be 3 g/h (5.38 sccm) and 30 sccm, respectively. The working pressure was in the range of 5–30 mTorr. The discharge power (P_{rf}) was varied from 30 to 150W by controlling the power generator and matching network.

Growth rate of the deposited film was measured in $\mu\text{m}/\text{min}$. The film thickness was measured using a profilometer (Tencor α step) from a step formed by shadowing a part of the silicon wafer with a polyimide foil during the deposition (the accuracy is ± 5 nm). The incorporation rate of Si—O—Si bridges into the films was estimated by measuring infra-red absorption spectra of the deposited films using a Fourier transformation infra-red spectrometer (FTIR, Perkin–Elmer 1800). Only the rocking mode around 450 cm^{-1} was used to determine the incorporation rate of Si—O—Si bridges, since the two other bands for Si—O—Si bridges are not distinguished clearly from other absorption bands due to overlapping [11]. The absorption intensity multiplied by the absorption cross-section for the bond gives the number of the bond in the films. Since the absorption cross-section is not known for that bond, the absorption intensity divided by the deposition time then gives the relative incorporation rate of the bond.

3. FTIR absorption spectra

Fig. 1 shows FTIR absorption spectra taken for all the films grown at $P_{rf} = 30$ W, but with different substrate temperatures, T_s , and oxygen flow rates, J_{O_2} : (a), $J_{O_2} = 0$ sccm and $T_s = \text{room temperature}$; (b), $J_{O_2} = 30$ sccm and $T_s = \text{room temperature}$; and (c), $J_{O_2} = 30$ sccm and $T_s = 200^\circ\text{C}$. All the spectra were normalized to a film thickness of 1 μm for comparison. In spectrum (a), two absorption bands are associated with CH bond groups, CH stretching around 3050–2810 cm^{-1} and CH deformation around 1520–1280 cm^{-1} [14], and three absorption bands are due to the three vibration modes of Si—O—Si, stretching

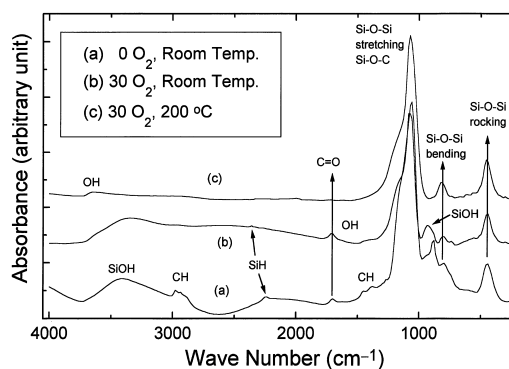


Fig. 1. IR absorption spectra for the films grown at $P_{rf} = 30$ W: (a), $J_{O_2} = 0$ sccm and $T_s = \text{room temperature}$; (b), $J_{O_2} = 30$ sccm and $T_s = \text{room temperature}$; and (c), $J_{O_2} = 30$ sccm and $T_s = 200^\circ\text{C}$.

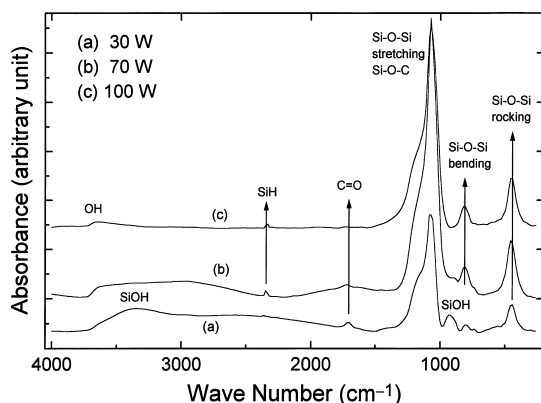


Fig. 2. IR absorption spectra for the films grown at $P_{rf} = 30, 70$ and 100 W with $J_{O_2} = 30$ sccm and $T_S =$ room temperature.

mode around 1090 cm^{-1} (or due to Si—O—C bonds [5]), bending mode around 820 cm^{-1} and rocking mode around 450 cm^{-1} . In addition to these bands, absorption bands due to SiOH around 960 and 3650 cm^{-1} [5], bonded OH around 3400 cm^{-1} [5], SiH around 2200 cm^{-1} [13], and C=O around 1720 cm^{-1} [5] are also present. From all these absorption bands, we see that some bonds such as Si—O (or Si—O—C) and CH in the monomer are preserved to be incorporated into films and some bonds are broken in the discharge to create new bond groups such as SiH, OH, and C=O. Spectrum (b) reveals that most organic groups are consumed in the presence of oxygen flow, the intensity of absorption due to OH, however, remains unchanged or even seems to have increased. It is further noted from spectrum (c) that the deposition at $T_S = 200^\circ\text{C}$, in the presence of oxygen flow, yields films containing less OH groups. Consequently, oxygen in the discharge leads to the formation of SiOH groups by oxidation of organic groups of the monomer molecule, and the SiOH groups are cross-linked to Si—O—Si bridges by thermally-activated processes.

Fig. 2 shows FTIR absorption spectra obtained for the films grown at $P_{rf} = 30, 70$ and 100 W, with $J_{O_2} = 30$ sccm and $T_S =$ room temperature. These spectra were also normalized to a film thickness of $1\text{ }\mu\text{m}$ for comparison. In these spectra, three absorption bands due to the three vibration modes of Si—O—Si are distinctively observable, being independent of P_{rf} . Other absorption bands associated with CH found in the films deposited without the oxygen flow are hardly observed for all the spectra. One of the systematic changes is the decrease in the absorption intensity around 3400 and 960 cm^{-1} associated with bonded OH and SiOH [5] with an increase in the discharge power. The absorption band due to C=O also decreases with P_{rf} . The absorption intensity due to Si—O—Si rocking around 450 cm^{-1} , however, does not seem to follow such a trend. This behavior is different from that for HMDSN/ O_2 discharges [12], and seems to be related to the dissociation of Si—O bonds of the monomer molecule at higher discharge powers. The presence of a band around 2300 cm^{-1} at $P_{rf} = 70$ and 100

W reflects the dissociation of Si—O bonds, since this band is attributed to SiH bond groups. The wave number for this band is somewhat higher than 2135 cm^{-1} for SiH stretching measured for amorphous films prepared in HMDSN discharges [13]. For films prepared in HMDSO discharges in the same condition, however, the wave number is around 2150 cm^{-1} , and increased systematically to about 2300 cm^{-1} [15] as the flow rate of oxygen gas increased from 0 to 50 sccm. This is well consistent with the finding by Lucovsky [16] that the frequency of the bond stretching vibration of SiH in amorphous solids varies with the electronegativity of the next nearest neighbor atoms of the network. Consequently, high discharge power yields increased integrity of deposited films by decreasing the density of SiOH, but induces increased dissociation of Si—O bonds of the monomer molecule, eventually resulting in the decreased growth rate or incorporation of Si—O—Si bridges into the films.

4. Deposition model

4.1. Surface kinetics model

4.1.1. Two-dimensional lattice model

For the formulation of a surface kinetics model, we firstly introduced a two-dimensional lattice of a growing TEOS-SiO₂ film, as shown in Fig. 3, in which $(\text{—Si—O—Si—})_n$ chains normal to the surface of substrate are defined as backbone chains and ones parallel to the surface as side chains, and a group of side chains in the same level makes an atomic layer. Four top-most atomic layers are shown in Fig. 3. It is assumed that the terminals of backbone chains are all occupied by SiOH ($\equiv\text{SiOH}^B$) or SiR groups ($\equiv\text{SiR}^B$). The terminals occupied by radicals or SiH groups are ignored. In this scheme, the reactions at the terminal of backbone chains lead to the growth of $(\text{—Si—O—Si—})_n$ chains, namely film growth, while the reactions in the side chains lead to cross-linking of backbone chains. Hence, the backbone chains are only made of $(\text{—Si—O—Si—})_n$ chains,

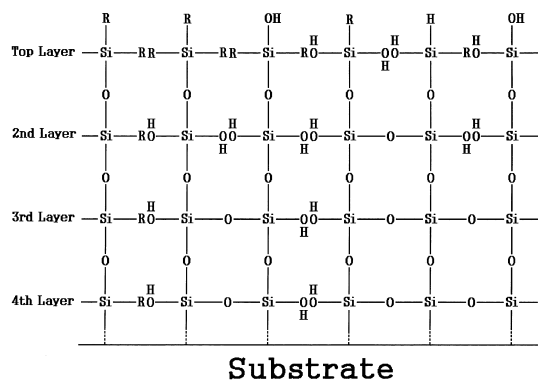


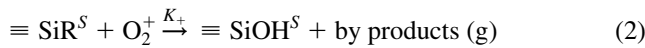
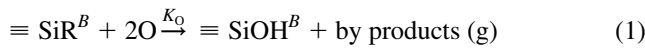
Fig. 3. A two-dimensional lattice of TEOS-SiO₂ films, in which $(\text{—Si—O—Si—})_n$ chains normal to the surface of substrate are defined as backbone chains and ones parallel to the surface as side chains.

the side chains contain some bonded OH ($\equiv\text{SiOH}^S$) and R groups ($\equiv\text{SiR}^S$), as shown in Fig. 3. Based on this two-dimensional lattice model, we propose a three-step reaction mechanism at the surface to film growth and cross-linking as follows.

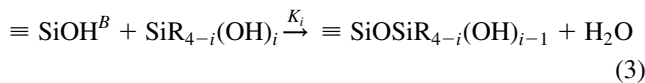
4.1.2. Reactions at the surface

4.1.2.1. First-step reactions

(reactions at the terminal of backbone chains leading to the formation of a new top layer) $\text{A}\equiv\text{SiR}^B$ is oxidized to $\text{a}\equiv\text{SiOH}^B$ by two oxygen atoms or one molecular oxygen ion coming to the surface of growing films, according to the chemical reaction model suggested by Sanogo and Zachariah [8] (Eqs. (1) and (2)).



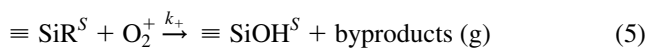
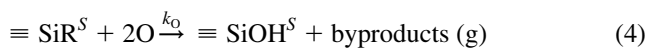
K represents the corresponding reaction rate constant and (g) represents gas state, respectively. Further, $\equiv\text{SiOH}^B$ is assumed only to react with $-\text{OH}$ groups of the precursors created from TEOS in the gas phase, to yield a $\text{Si}-\text{O}-\text{Si}$ bridge. We assume that the precursors in the discharge are all in the form of $\text{SiR}_{4-i}(\text{OH})_i$ with $i = 1-4$, the reason for which to be explained in section 4.2.



K_i is the reaction rate constant for Eq. (3). For $i = 1$, this reaction results in the loss of a terminal occupied by $\equiv\text{SiOH}^B$, yielding a $\text{a}\equiv\text{SiR}^B$ and two $\equiv\text{SiR}^S$. For $i = 2$, as a result of this reaction, we have either a $\text{a}\equiv\text{SiOH}^B$ and two $\equiv\text{SiR}^S$, or a $\text{a}\equiv\text{SiR}^B$, a $\text{a}\equiv\text{SiOH}^S$ and a $\text{a}\equiv\text{SiR}^S$. To simplify the modeling process, we only consider the case where $-\text{OH}$ in $\text{SiR}_{4-i}(\text{OH})_i$ is arranged into the backbone chain to be $\equiv\text{SiOH}^B$. We then have a $\text{a}\equiv\text{SiOH}^B$, a $\text{a}\equiv\text{SiOH}^S$ and a $\text{a}\equiv\text{SiR}^S$ for $i = 3$. Any reactions between $\equiv\text{SiR}^B$ and $-\text{OH}$ in $\text{SiR}_{4-i}(\text{OH})_i$, or $\equiv\text{SiOH}^B$ and $-\text{R}$ in $\text{SiR}_{4-i}(\text{OH})_i$ are ignored.

4.1.2.2. Second-step reactions

(reactions in the side chains of the old top layer) As a new top atomic layer is forming through Eqs. (1)–(3), the old top layer becomes the second layer. In this while, some of the $\equiv\text{SiR}^S$ of the old top layer is oxidized to $\equiv\text{SiOH}^S$ by incoming O or O_2^+ , as $\equiv\text{SiR}^B$ is oxidized to $\equiv\text{SiOH}^B$.



where k_0 and k_+ are the reaction rate constants for Eqs. (4) and (5), respectively. It is assumed that $\equiv\text{SiOH}^S$ of the top layer is unable to react with any $-\text{OH}$ in $\text{SiR}_{4-i}(\text{OH})_i$

coming to the surface. Some $\equiv\text{SiOH}^S$ react with other $\equiv\text{SiOH}^S$ in the next backbone chain to cross-link to $\text{Si}-\text{O}-\text{Si}$ bridges by dehydration.



where k_s is the reaction rate constant for Eq. (6). Any reactions between $\equiv\text{SiR}^S$ and $\equiv\text{SiOH}^S$ are ignored.

4.1.2.3. Third-step reactions

(reactions in the side chains of the old second layer) At the time of forming a new top layer, the old second layer becomes the third layer. It is assumed that the approach of O or O_2^+ to the old second layer is blocked by the old top layer, so that $\equiv\text{SiR}^S$ of the old second layer is not attacked by O or O_2^+ . Dehydration (Eq. (6)) is the only reaction in the side chain of the second layer.

The lower layers are assumed to be immune to any reactions. Some dehydration reaction can occur, but the evaporation of H_2O is hindered by the upper layers. The reaction rate is expected to be very low, and thus the reaction is ignored.

4.1.3. Rate of film growth

Let θ_{OH} be the fraction of $\equiv\text{SiOH}^B$, then in the steady state (Eq. (7))

$$\begin{aligned} \frac{d\theta_{\text{OH}}}{dt} &= -K_1 n_1 \theta_{\text{OH}} + K_0 n_0^2 (1 - \theta_{\text{OH}}) + K_+ n_+ (1 - \theta_{\text{OH}}) \\ &= 0 \end{aligned} \quad (7)$$

where n denotes the densities of the corresponding chemical species. The first term in the right-hand side of Eq. (7) is the loss of $\equiv\text{SiOH}^B$ due to Eq. (3). The second and third terms are the gain of $\equiv\text{SiOH}^B$ due to oxidation of $\equiv\text{SiR}^B$. Eq. (7) solves for θ_{OH} as in Eq. (8)

$$\theta_{\text{OH}} = \frac{K_0 n_0^2 + K_+ n_+}{K_0 n_0^2 + K_+ n_+ + K_1 n_1} \quad (8)$$

The film growth rate, D , is given in Eq. (9a) as

$$D = \frac{a_0}{n_G} \sum K_i n_i \theta_{\text{OH}} = \frac{a_0}{n_G} \frac{\sum K_i n_i (K_0 n_0^2 + K_+ n_+)}{K_0 n_0^2 + K_+ n_+ + K_1 n_1} \quad (9a)$$

where a_0 and n_G are the length of a $\text{Si}-\text{O}-\text{Si}$ bridge and the density of backbone chains, respectively. In the scheme where the ratio of O_2 to TEOS flow is very high, as is often the case, $K_0 n_0^2 + K_+ n_+ \gg K_1 n_1$, Eq. (9a) reduces to Eq. (9b)

$$D = \frac{a_0}{n_G} \sum K_i n_i \quad (9b)$$

Hence, the deposition model of this work tells us that D is only governed by the film precursors generated from TEOS in the gas phase, if the ratio of O_2 to TEOS flow is very high.

4.1.4. Incorporation of $\equiv\text{SiOH}^S$ and $\equiv\text{SiR}^S$

We let ξ_i^R and ξ_i^{OH} be the fractions of $\equiv\text{SiR}^S$ and $\equiv\text{SiOH}^S$ in the top ($i = 1$), second ($i = 2$) and third layer ($i = 3$), respectively. As a result of Eq. (3) in the first-step, we have 2, 2, 1 and 0 $\equiv\text{SiR}^S$ for $i = 1, 2, 3$ and 4, respectively. ξ_i^R and ξ_i^{OH} are then given as (see Eqs. (10) and (11))

$$\xi_1^R = \frac{2K_1n_1 + 2K_2n_2 + K_3n_3}{2\sum K_in_i} \quad (10)$$

$$\xi_1^{\text{OH}} = 1 - \xi_1^R = \frac{K_3n_3 + 2K_4n_4}{2\sum K_in_i} \quad (11)$$

As a result of Eqs. (4) and (5) in the second-step, a fraction of ξ_1^R becomes ξ_2^{OH} . A fraction of ξ_1^{OH} is cross-linked to Si—O—Si bridges via Eq. (6). Resultantly, in Eqs. (12) and (13)

$$\xi_2^R = \xi_1^R \{1 - (k_0n_0^2 + k_+n_+)\} \quad (12)$$

$$\xi_2^{\text{OH}} = \xi_1^R(k_0n_0^2 + k_+n_+) + \xi_1^{\text{OH}}(1 - 2k_S) \quad (13)$$

In the third-step reactions, no $\equiv\text{SiR}^S$ is oxidized, and a fraction of ξ_2^{OH} is dehydrated via Eq. (6). Therefore, (see Eqs. (14) and (15))

$$\xi_3^R = \xi_2^R \quad (14)$$

$$\xi_3^{\text{OH}} = \xi_2^{\text{OH}}(1 - 2k_S) \quad (15)$$

Solving Eqs. (10)–(15) for ξ_3^R and ξ_3^{OH} yields Eqs. (16a) and (17a)

$$\xi_3^R = \frac{2K_1n_1 + 2K_2n_2 + K_3n_3}{2\sum K_in_i} \{1 - (k_0n_0^2 + k_+n_+)\} \quad (16a)$$

$$\xi_3^{\text{OH}} = \frac{(1 - 2K_S)}{2\sum K_in_i} \{(K_3n_3 + 2K_4n_4)(1 - 2k_S) + (2K_1n_1 + 2K_2n_2 + K_3n_3)(k_0n_0^2 + k_+n_+)\} \quad (17a)$$

In the case of the negligible densities of $\text{SiR}(\text{OH})_3$ and $\text{Si}(\text{OH})_4$, it holds, $K_1n_1 + K_2n_2 \gg K_3n_3 + K_4n_4$. Eqs. (16a) and (17a) reduce to Eqs. (16b) and (17b), respectively

$$\xi_3^R = 1 - (k_0n_0^2 + k_+n_+) \quad (16b)$$

$$\xi_3^{\text{OH}} = (1 - 2k_S)(k_0n_0^2 + k_+n_+) \quad (17b)$$

Hence, the fraction of cross-linking, i.e. the fraction ξ_G of Si—O—Si bridges in the side chain of bulk film is given as the following Eq. (18)

$$2\xi_G = 1 - \xi_3^{\text{OH}} - \xi_3^R = 2k_S(k_0n_0^2 + k_+n_+) \quad (18)$$

The incorporation rate ψ of Si—O—Si bridges into growing films, which is a measurable property, is written as Eq. (19a)

$$\psi = \sum K_in_i\theta_{\text{OH}}(1 + 2\xi_G) \quad (19a)$$

or, from Eq. (9a) as Eq. (19b)

$$\psi = \frac{n_G D}{a_0}(1 + 2\xi_G) \quad (19b)$$

Inserting Eq. (18) into Eq. (19b) yields Eq. (20)

$$\psi = \frac{n_G D}{a_0}(1 + 2k_S(K_0n_0^2 + k_+n_+)) \quad (20)$$

Accordingly, the ratio ψ/D is only determined by the surface reactions associated with oxygen radicals and molecular ions and by the dehydration process.

4.2. Density of chemical species in the steady state

We give here the densities of chemical species in the steady state of plasma as a function of the discharge power, based on the dissociation model proposed in our previous work [11], and relate them to the rate of film growth and incorporation of Si—O—Si bridges derived from the surface kinetics model of this work.

As most of the bond energies of the monomer molecule are lower than a few eV, almost all of the bonds can be dissociated by electron impact. According to the calculation of Ho and Melius [17], the dissociation of TEOS by the breaking of an O—C bond and subsequent formation of SiR_3OH , i.e. as in Eq. (21),



has the lowest enthalpy (10.5 kcal/mol or 0.46 eV) among other possible dissociation reactions. Further dissociation of SiR_3OH to $\text{SiR}_2(\text{OH})_2$, $\text{SiR}_2(\text{OH})_2$ to $\text{SiR}(\text{OH})_3$, and $\text{SiR}(\text{OH})_3$ to $\text{Si}(\text{OH})_4$ are also energetically most favorable with dissociation enthalpies around 10 kcal/mol [17]. If the dissociation of TEOS to $\text{Si}(\text{OH})_4$ proceeds in a consecutive manner, the process can be summarized in Eq. (22) as follows



where P_i stands for $\text{SiR}_{4-i}(\text{OH})_i$, and k_i is the reaction rate constant for each individual dissociation reaction. According to the model of our previous work [11], the density, n_i , of P_i generated via Eq. (22) is given, from the consideration of material balance in the steady state, as Eq. (23)

$$k_in_{i-1} = \frac{n_i}{\tau_i} + k_{i+1}n_i \quad (23)$$

where τ_i is the averaged residence time of P_i in the plasma zone and assumed to be a weak function of, or invariant under, the variation of P_{rf} . Where $\tau_i = 0$, this means some direct dissociation of TEOS to P_2 , or P_1 to P_3 , etc. As τ_i is the averaged residence time, Eq. (23) comprises even such cases. The term in the left-hand side of Eq. (23) is the rate of increase of P_i via dissociation of the precursor P_{i-1} . The first and second terms in the right-hand side of Eq. (23) refer to the rate of decrease by out-flow and due to dissociation of P_i to P_{i+1} , respectively. k_i is related to the discharge power P_{rf} with a parameter η_i as $k_i \equiv \eta_i P_{\text{rf}}$ [11]. For $i = 0$, k_in_{i-1} refers to q_M , the monomer input rate per volume of plasma. The

density, n_0 , of the monomer molecule in the steady state, is then given as Eq. (24) [11]

$$n_0 = \frac{q_M}{\eta_1 P_{rf} + \tau_0^{-1}} = \frac{q_M}{\eta_1 (P_{rf} + \chi_1^{-1})} \quad (24)$$

with $\chi_1 = \eta_1 \tau_0$, the dissociation efficiency for reaction (21) per watt of input. n_1 and n_2 are further given from Eq. (23) as Eqs. (25) and (26)

$$n_1 = \frac{n_0 \eta_1 P_{rf}}{\eta_2 P_{rf} + \tau_1^{-1}} = \frac{q_M P_{rf}}{\eta_2 (P_{rf} + \chi_1^{-1})(P_{rf} + \chi_2^{-1})} \quad (25)$$

$$n_2 = \frac{n_1 \eta_2 P_{rf}}{\eta_3 P_{rf} + \tau_2^{-1}} = \frac{q_M P_{rf}}{\eta_3 (P_{rf} + \chi_1^{-1})(P_{rf} + \chi_2^{-1})(P_{rf} + \chi_3^{-1})} \quad (26)$$

with $\chi_i = \eta_i \tau_{i-1}$. n_3 and n_4 can be derived in the same way. Ignoring them as insignificant, we have the growth rate from Eq. (9b) as shown in Eq. (27a)

$$D = \frac{a_0 q_M P_{rf}}{n_G (P_{rf} + \chi_1^{-1})(P_{rf} + \chi_2^{-1})} \left(\frac{K_1}{\eta_2} + \frac{K_2}{\eta_3} \frac{P_{rf}}{P_{rf} + \chi_3^{-1}} \right) \quad (27a)$$

If we ignore loss of P_2 due to the generation of P_3 , letting $\eta_3 P_{rf} \approx 0$ in Eq. (26), Eq. (27a) reduces to that shown in Eq. (27b)

$$D = \frac{\alpha P_{rf} (P_{rf} + \gamma^{-1})}{(P_{rf} + \chi_1^{-1})(P_{rf} + \chi_2^{-1})} \quad (27b)$$

with (Eq. (28))

$$\alpha = \frac{a_0 q_M \tau_2 K_2}{n_G} \quad (28a)$$

and

$$\lambda = \frac{\eta_2 \tau_2 K_2}{K_1} \quad (28b)$$

The density of oxygen atoms, n_O , is derived in the same way and given as [12]

$$n_O = \frac{2q_{O_2} \tau_O P_{rf}}{P_{rf} + \chi_O^{-1}} \quad (29)$$

All the parameters in Eq. (29) have the same meaning as those in Eq. (24) but for oxygen atoms and molecules. The molecular oxygen ion density, n_+ , can be written in the same form as in Eq. (29). Reproducing the data of ion density measured for Ar/O₂ plasmas [9] as the reverse ion density vs. the reverse discharge power, indeed yields straight lines, being independent of oxygen percent in Ar, as shown in Fig. 4. This result renders n_+ to be expressed as (Eq. (30))

$$n_+ = \frac{q_{O_2} \tau_+ P_{rf}}{P_{rf} + \chi_+^{-1}} \quad (30)$$

with τ_+ and χ_+ denoting the residence time of the ion and the efficiency of ionization per watt of input, respectively. The value of χ_+^{-1} is $\sim 10^4$ W, estimated from Fig. 4, whereas

that of χ_O^{-1} is $\sim 10^2$ W, estimated using the data in the same reference. Hence, $\chi_+^{-1} \gg P_{rf}$ for the range of discharge power used in this work, and Eq. (30) is approximated to $n_+ = \chi_+ \tau_+ q_{O_2} P_{rf}$. Inserting Eqs. (29) and (30) into Eq. (18) yields a form for ξ_G in Eq. (31)

$$\xi_G = k_S k_O \left(\frac{2\tau_O q_{O_2} P_{rf}}{P_{rf} + \chi_O^{-1}} \right)^2 + k_S k_+ \chi_+ \tau_+ q_{O_2} P_{rf} \quad (31)$$

In Eq. (6), we did not differentiate the thermally-activated process of dehydration from the ion-induced. Considering that the thermally-activated process is supported by ion bombardment, k_S can be described by the combination of these two factors. At high P_{rf} , where ions play a dominant role, k_S represents only the effect by ion bombardment. At low P_{rf} , the density of O₂⁺ is low, and thus the second term in the right-hand side of Eq. (31) is negligible. Consequently, the second term refers only to the effect of ions.

5. Comparison of the model with the experimental results

5.1. Film growth rate

Fig. 5 shows the film growth rate for two different T_S as a function of P_{rf} . It is noted that the growth rate for higher T_S is lower at lower P_{rf} , but higher at higher P_{rf} , and the rate shows a maximum around 50 and 80 W for the films deposited at room temperature and 200°C, respectively. The decrease in the growth rate at higher P_{rf} seems to be associated with the increased dissociation of Si—O bonds of the monomer molecule, as was noted from the observation of FTIR absorption spectra in section 3. We first, however, interpret this behavior based on Eq. (27b), being derived without considering the dissociation of Si—O bonds. The condition that Eq. (27b) has a maximum in terms of P_{rf} is $dD/dP_{rf} = 0$ with $P_{rf} > 0$, so it must hold (Eq. (32))

$$\frac{K_1}{K_2} > \frac{\eta_2 \tau_2}{\eta_1 \tau_0} + \frac{\tau_2}{\tau_1} \quad (32)$$

As no values for the parameters in Eq. (32) are known, the

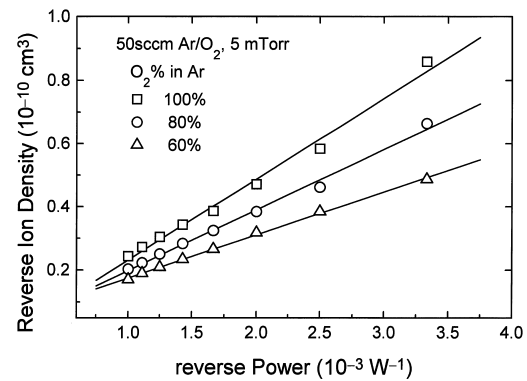


Fig. 4. The measured data on ion densities of Ar/O₂ plasmas reported in [9] as the reverse ion density vs. the reverse discharge power.

inequality can not be verified. Nevertheless, the data in Fig. 5 are fitted based on Eq. (27b) by letting $\chi_1 \equiv \chi_2 = \chi_{\text{O-C}}$, the efficiency of dissociation of O–C bonds of the monomer molecule. The results are shown in the same figure with two dot lines. The value for $\chi_{\text{O-C}}$ is estimated to be about $2 \times 10^{-2} \text{ W}^{-1}$ for the two fitted lines, well comparable to those obtained for the dissociation of Si–O–Si ($2.5 \times 10^{-2} \text{ W}^{-1}$), Si–C ($5.8 \times 10^{-2} \text{ W}^{-1}$), and C–H ($2.2 \times 10^{-2} \text{ W}^{-1}$) bonds in HMDSO plasmas [11]. The data for $T_s = 200^\circ\text{C}$ are fitted well to the line, but not well for $T_s = \text{room temperature}$. This discrepancy seems to be due to the fact that we have ignored the effect of the dissociation of the Si–O bond of the monomer molecule on the calculation. According to Ho and Melius [17], the dissociation of Si–O bond and the accompanying generation of the SiH group is energetically most favorable except for the reaction in Eq. (21). The calculated dissociation enthalpy of the following reaction in Eq. (33) is 45.8 kcal/mol



If we take Eq. (33) into account, we have an additional set of equations for dissociation and surface reaction to obtain a more complicated formula. For simplicity, however, we assume that R_3SiH generated via Eq. (33) does not participate in the process of film growth. The density of the monomer molecule, n_M , survived from Eq. (33) is given as the following in Eq. (34) [11]

$$n_M = \frac{q_M \tau_0}{\chi_{\text{Si-O}} P_{\text{rf}} + 1} \quad (34)$$

with $\chi_{\text{Si-O}}$ the dissociation efficiency for Eq. (33). The modified growth rate D_m is thus (Eq. (35))

$$D_m = \frac{D}{\chi_{\text{Si-O}} P_{\text{rf}} + 1} \quad (35)$$

The data in Fig. 5 are fitted again based on Eq. (35), the results of which are shown in the same figure with two solid lines. For the fitting process, we used the value for $\chi_{\text{O-C}}$, $2 \times 10^{-2} \text{ W}^{-1}$, estimated based on Eq. (27b), and that for $\chi_{\text{Si-O}}$, $0.46 \times 10^{-2} \text{ W}^{-1}$, calculated from the ratio of dissociation

enthalpies for the reactions in Eqs. (21) and (33). The data for $T_s = \text{room temperature}$ is fitted better with Eq. (35), while for $T_s = 200^\circ\text{C}$ the difference is trivial. As a result of the fitting process, the values of α are estimated to be 0.21 and $0.75 \text{ } \mu\text{m}/\text{min}$, and that of γ to be 0.22×10^{-2} and $1.29 \times 10^{-2} \text{ W}^{-1}$ for $T_s = \text{room temperature}$ and 200°C , respectively. It is well established that with increasing T_s the growth rate decreases, resulting in negative activation energies for film growth [2,5,10,18]. The negative activation energy is mostly taken as an indication of the adsorption reaction of reactive species at the surface [5,10,18]. The reverse trend at the higher P_{rf} of this work, however, cannot be explained consistently in this way. Using the model of this work and estimated values of α and γ , the overall trend can be interpreted.

As the parameters χ in Eqs. (27b) or (35) do not depend on T_s , the difference in D associated with the difference in T_s at low powers is determined by the difference in the ratio, α/γ , according to Eq. (27b). The estimated ratio is higher for $T_s = \text{room temperature}$, leading to higher growth rate. At enough high powers, however, the difference is only determined by the difference in α , according to Eq. (27b). The estimated value of α is higher for $T_s = 200^\circ\text{C}$, hence, the growth rate is higher. Therefore, negative activation energy is yielded when the deposition is carried out in the low discharge power regime. According to Eq. (28), α/γ gives $a_0 q_M K_1 / n_G \eta_2$, where only K_1 is temperature-dependent, whereas in α only K_2 is temperature-dependent. According to our previous work on the calculation of apparent activation energy for film deposition of TEOS-SiO₂ [19], the activation energy is determined by two terms. One is the activation energy for surface reaction of gaseous species, being positive, and the other is due to the temperature-dependent distribution of gaseous species in the volume of an isobaric plasma, leading to a negative activation energy. It is concluded, therefore, that K_1 is more dependent on the temperature-dependent distribution of gaseous species than K_2 , and that the surface activation energy for K_2 is more positive than that due to the distribution is negative.

5.2. Incorporation rate of Si–O–Si bridges

Fig. 6 shows the value of ψ as a function of P_{rf} . ψ for $T_s = 200^\circ\text{C}$ increases steadily, whereas that for $T_s = \text{room temperature}$ shows a maximum. It is also noted that ψ for $T_s = 200^\circ\text{C}$ is higher than that for $T_s = \text{room temperature}$ over the entire range of P_{rf} studied in this work, which makes a clear difference from the growth rate shown in Fig. 5. This difference is due to the additional effect by gaseous oxygen atoms and molecular ions leading to cross-linking of deposited films, according to the model of this work described in Eq. (20). The ratio, ψ/D , which gives the relative density of Si–O–Si bridges, $1 + 2\xi_G$, reflects this additional effect. Fig. 7 shows the ratio as a function of P_{rf} . Steady increase in the ratio is noted for both T_s 's, and the ratio is higher for $T_s = 200^\circ\text{C}$. The difference between the

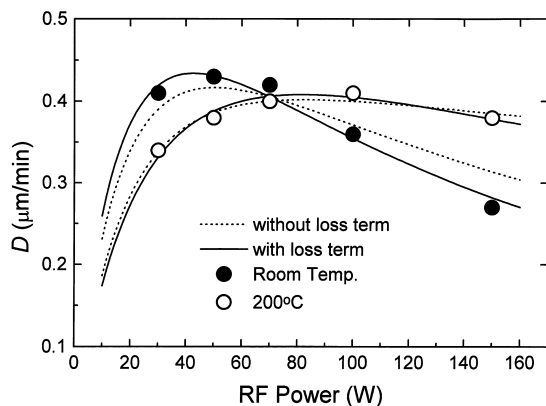


Fig. 5. The growth rate D for the films grown at $T_s = 200^\circ\text{C}$ and room temperature as a function of P_{rf} .

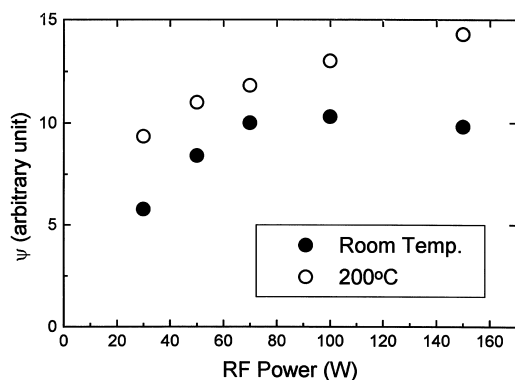


Fig. 6. The incorporation rate ψ of the Si—O—Si bridge for the films grown at $T_S = 200^\circ\text{C}$ and room temperature as a function of P_{rf} .

two T_S 's, however, becomes lower with increasing P_{rf} . The behavior of the ratio as a function of P_{rf} can be explained in terms of Eqs. (20) and (31). At enough low discharge powers, any ion-induced effects are weak, and thus, the second term in the right-hand side of Eq. (31) can be ignored. Then, $k_S k_O$ in the first term is responsible for the difference. The activation energy for the intramolecular elimination of H_2O such as that in Eq. (6) is reported be 15 000 K (1.293 eV) [9], while that for the reaction of TEOS with O is 1304 K (0.112 eV) [8]. The difference in k_S seems to give rise to an apparent difference in the ratio between the two T_S 's. At enough high discharge powers the effect by ions becomes dominant. As ion-induced reactions are independent of T_S , as was noted by Raupp et al. [10], the density of cross-linking increases linearly and the difference in the ratio becomes smaller with increasing P_{rf} from Eq. (31), consistent with the experimental results shown in Fig. 7. It should be noted here that the value of $2\xi_G$ ranges from zero to one, as defined in Eq. (18). The ratio shown in arbitrary unit in Fig. 7 goes up to 40 or higher with increasing P_{rf} . If we take 40 to be the maximum in arbitrary unit associated with the maximum value of $1 + 2\xi_G$ in Eq. (19b), then the minimum ratio in arbitrary unit should be 20 for $\xi_G = 0$. The ratio for $P_{\text{rf}} < 50$ W and $T_S = \text{room temperature}$, is, however, lower than 20. This indicates that backbone chains in the films obtained bridges at low discharge powers and room temperature contain less Si—O—Si bridges than predicted by Eq. (9b), and thus that other bonds such as Si—O—C or some broken backbone chains in the form of —SiOH/OHSi— are incorporated into the films, eventually leading to a gel-like material [4].

6. Summary

TEOS based SiO_2 films were deposited on silicon wafers in an RF reactor at various discharge powers and two substrate temperatures at a TEOS flow rate of 5.38 sccm. Infra-red absorption spectra were taken for each deposited film using an FTIR spectrometer. FTIR spectra for the films

deposited at a discharge power of 30 W in the absence of oxygen flow show a strong absorption intensity for Si—O—Si and Si—OH bonds along with CH bond groups. In the presence of an oxygen flow of 30 sccm, the absorption intensity due to CH decreases to a very low level, while the intensity due to Si—O—Si and SiOH remains unchanged or increased. Heating the substrate to a temperature of 200°C leads to a fair decrease in the intensity due to SiOH, leaving the film more oxide-like. Increasing the discharge power gives rise to a decrease in the intensity due to Si—OH, but induces SiH groups into the films.

Based on these observations, a model for the plasma and surface kinetics was proposed. Using a proposed three-step reaction mechanism at the surface of a two-dimensional lattice, the growth rate and the incorporation rate of Si—O—Si bridges into the films were derived as functions of the densities of the film precursors, oxygen atoms and molecular oxygen ions. The densities are given by the uniform discharge model of our previous work. According to the model of this work, in the regime of a high ratio of oxygen to TEOS flow the growth rate is only governed by the film precursors generated from TEOS and their reactions with surface, while the incorporation rate is additionally determined by the surface reactions associated with oxygen radicals and ions and by the dehydration process.

The measured growth rate shows a maximum for the films deposited either at 200°C or room temperature, but is higher for the films deposited at room temperature for low discharge powers, and lower for high discharge powers than those deposited at 200°C . The growth rate, as derived for the scheme of dominant film precursors of SiR_3OH and $\text{SiR}_2(\text{OH})_2$, shows a good agreement with the experimental results. The dissociation efficiency for O—C and Si—O bonds of the monomer molecule are estimated to be about 2×10^{-2} and $0.46 \times 10^{-2} \text{ W}^{-1}$, respectively. Compared to the growth rate, the incorporation rate is higher for the films deposited at 200°C than that for those deposited at room temperature in the whole range of discharge power studied in this work, which resulted from the additional effect by oxygen atom or ions. The ratio of the two rates, which reflects this additional effect, gives the relative density of

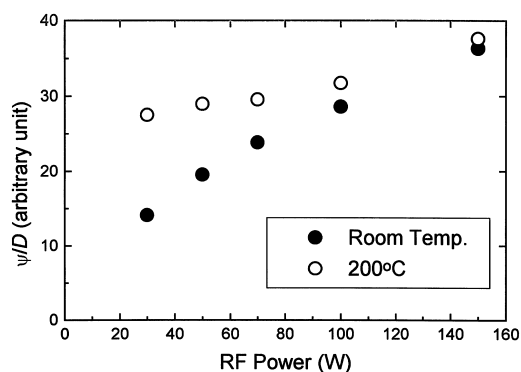


Fig. 7. The ratio of ψ to D as a function of P_{rf} .

Si—O—Si bridges in the films and shows a dominant effect of substrate heating to 200°C at low discharge powers leading to a higher density of Si—O—Si bridges in the films. The effect, however, becomes smaller with increasing discharge power, as the role of ions becomes more dominant, according to the proposed model of this work.

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